

Topic quiz: Forces

IBDP Physics SL · Grade 11 · A.2 Forces · Newton's laws · 10 multiple-choice questions

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Name: _____

Class: _____

Date: _____

How to use. Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use $g = 9.81 \text{ m s}^{-2}$ and show any working next to the question.

Check yourself online: the same quiz with instant feedback and explanations —
<https://biba0112.github.io/physics-sims/quiz-forces/>



1	2	3	4	5	6	7	8	9	10

Question 1

A book rests on a table. According to Newton's third law, which force is the partner of the book's weight?

- ☐ A The normal force of the table on the book
- ☐ B The push of the book on the table
- ☐ C The gravitational pull of the book on the Earth
- ☐ D The friction of the table on the book

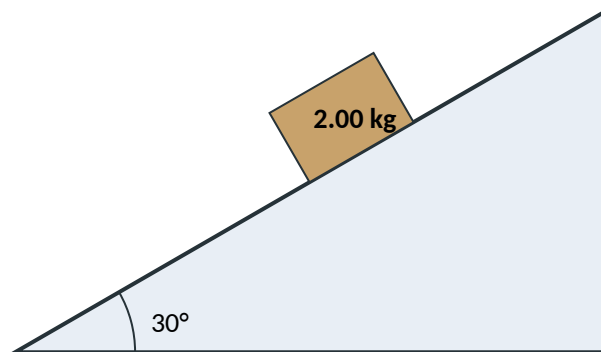
Question 2

A 60.0 kg student stands on bathroom scales in a lift that is accelerating upwards at 1.50 m s^{-2} . What do the scales read?

- ☐ A 90 N
- ☐ B 499 N
- ☐ C 589 N
- ☐ D 679 N

Question 3

A 2.00 kg block rests on a ramp at 30° to the horizontal. What is the normal force on the block?



(A) 9.81 N

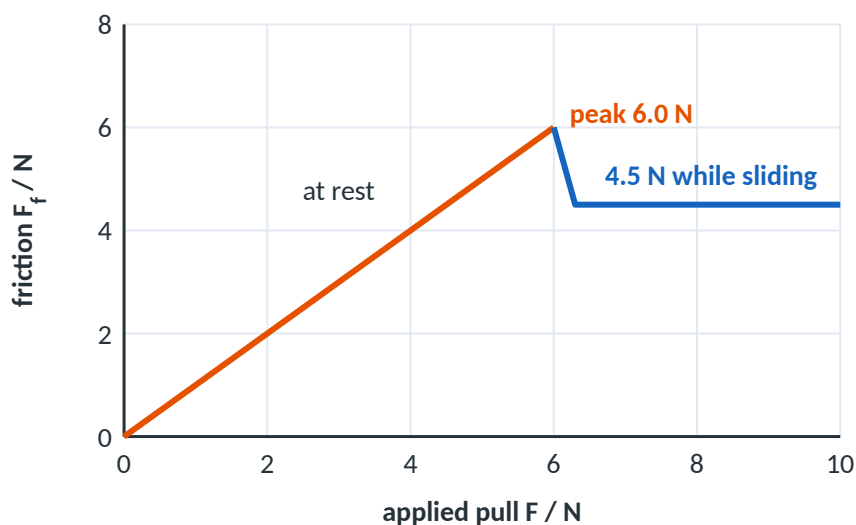
(B) 17.0 N

(C) 19.6 N

(D) 22.7 N

Question 4

A block of weight 15.0 N is pulled along a horizontal bench with a slowly increasing force. The graph shows the friction force. What are the coefficients of static and dynamic friction?



(A) $\mu_s = 0.30$ and $\mu_d = 0.40$

(B) $\mu_s = 0.40$ and $\mu_d = 0.40$

(C) $\mu_s = 0.40$ and $\mu_d = 0.30$

(D) $\mu_s = 2.5$ and $\mu_d = 3.3$

Question 5

A ramp is tilted slowly. A block on it just starts to slip when the angle is 28° . What is the coefficient of static friction?

(A) 0.47

(B) 0.53

(C) 0.88

(D) 1.9

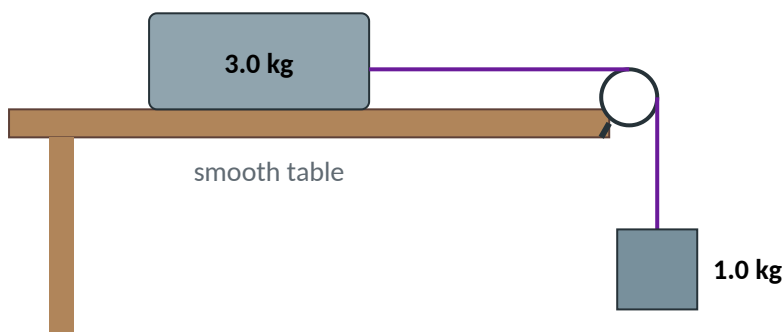
Question 6

The block from the last question is replaced by a heavier block of the same material. At what angle does the heavier block start to slip?

- (A) At a smaller angle, because its weight down the slope is larger
- (B) At a larger angle, because the friction force is larger
- (C) At the same angle, because the mass cancels out
- (D) It never slips

Question 7

A 3.0 kg cart on a smooth table is pulled by a string over a pulley by a hanging 1.0 kg mass. What is the tension in the string while the masses accelerate?



- (A) 2.45 N
- (B) 7.36 N
- (C) 9.81 N
- (D) 29.4 N

Question 8

A spring stretches by 4.0 cm when a 0.20 kg mass hangs from it at rest. What is the spring constant?

- (A) 0.049 N m^{-1}
- (B) 0.49 N m^{-1}
- (C) 5.0 N m^{-1}
- (D) 49 N m^{-1}

Question 9

A steel ball falls through glycerol at its terminal velocity. W is its weight, F_d the drag and F_b the buoyant force. Which equation is correct?

- (A) $W = F_d + F_b$
- (B) $F_d = W + F_b$
- (C) $F_d = 0$
- (D) $W > F_d + F_b$

Question 10

An object weighs 2.50 N in air and 1.88 N when fully under water ($\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$). What is its volume?

- (A) 63 cm^3
- (B) 192 cm^3
- (C) 255 cm^3
- (D) 620 cm^3